Generalized Nonconvex Low-Rank Approximation and Sparse Regularizer for Hyperspectral Image Denoising

Yusen Tan, Dong Hu, Yao Fu, Chunming Yang, and Zhi Wang, *Member, IEEE*

***Abstract*—Hyperspectral images (HSIs) are frequently corrupted by mixed sparse and Gaussian noise during acquisition and transmission processes, which degrades image quality and compromises subsequent analysis. In recent years, low-rank approximation-based methods have achieved promising results in HSI denoising and have garnered widespread attention. However, most of these methods often lack robustness, failing to deliver consistent performance across diverse noise scenarios, and tend to oversmooth fine image details. To address these issues, this article fully explores low-rank and sparse prior knowledge of HSIs and proposes a novel generalized nonconvex framework for HSI denoising. The proposed method offers two significant advantages. First, the model exhibits high flexibility. By selecting scenario-specific nonconvex surrogates, its generalized nonconvex surrogate ensures optimal singular value recovery performance of HSIs across diverse noise scenarios. Second, by enhancing the distinction between image details and noise, its sparse regularizer further improves singular value recovery accuracy of HSIs. To solve the proposed model, an efficient algorithm with theoretical guarantees is developed by leveraging the well-known inexact augmented Lagrange multiplier (ALM) framework. Experimental results on both simulated and real datasets demonstrate that the proposed method outperforms state-of-the-art approaches and achieves superior results both quantitatively and visually. MATLAB code is available at https://github.com/wangzhi-swu/GNRSR**

***Index Terms*—Augmented Lagrange multiplier (ALM), denoising, generalized nonconvex low-rank framework, hyperspectral image (HSI), sparsity.**

I. Introduction

HYPERSPECTRAL images (HSIs) are acquired by satellites or imaging spectrometers with hundreds or thousands of narrow spectral bands, which provide abundant spatial information and spectral features. In recent years, HSIs have garnered significant attention across various fields, including environmental monitoring [1], disease diagnosis [2], and crop classification [3], due to their exceptional capability for material identification. Unfortunately, because of the influence of sensor sensitivity and the complex natural environment, the quality of captured HSIs is often degraded by mixed noise, such as Gaussian noise, deadlines, stripes, and impulse noise. The presence of noise seriously affects the quality of subsequent applications, such as anomaly detection [4], unmixing [5], and classification [6]. Therefore, effective denoising of HSIs has become an essential step prior to their application.

During the past decades, numerous methods have been proposed for HSI denoising. From a structural point of view, HSIs can be regarded as a stack of multiple grayscale images. Therefore, an intuitive method is to treat each spectral band of the HSIs as an independent grayscale image and to denoise band-by-band. Accordingly, the conventional methods such as NL-means [7], K-SVD [8], and BM3D [9] can be employed for the restoration of HSIs. Unfortunately, the computing cost of NL-means and K-SVD is high [9], [10], and BM3D cannot handle some sparse image blocks [11]. Taking advantage of SVD to finely discriminate similar and dissimilar pixel pairs, and then remove dissimilar pixel pairs, Orchard et al. [10] proposed a more efficient method than NL-means, which can achieve better denoising effects. Protter and Elad [12] propagated the dictionary from frame to frame to reduce the number of iterations, and thus proposed a low computational complexity image sequence denoising method based on K-SVD. By combining shape adaptive transform and principal component analysis, Dabov et al. [11] have introduced BM3D-SAPCA that enhances sparse representation capabilities. However, these methods fail to fully exploit the strong spectral correlation between bands, as well as the spatial continuity within each band, which results in limited accuracy in recovering both image details and spectral characteristics.

HSIs encompass rich spectral features and spatial information. Specifically, spectral features have different spectral reflection information in different image scenes, helping to distinguish the noise and signal components [13]. Meanwhile, spatial information includes the texture, edge, spatial structure, and other features of the image, which helps to retain more details of the image during the denoising process [14]. Therefore, the joint utilization of spectral–spatial information helps to improve the recovery accuracy of HSIs. To take full advantage of this information, researchers have proposed a series of improved approaches, including SSAHTV [14], SSTV [15], and GSSTV [16]. Yuan et al. [14] developed the SSAHTV model by considering the differences in noise intensity and

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Yusen Tan, Dong Hu, Yao Fu, and Zhi Wang are with the College of Computer and Information Science, Southwest University, Chongqing 400715, China (e-mail: chiw@swu.edu.cn).

Chunming Yang is with the School of Computer Science and Technology, Southwest University of Science and Technology, Mianyang 621010, China.

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spatial information between different bands. The SSAHTV model preserves edge information well, but, unfortunately, the denoising result is too smooth and loses a lot of local details [17]. To alleviate the problem, Yuan et al. [17] have introduced the spectral–spatial kernel model by leveraging the inherent regularity of local structures and the self-similarity within spectral, which can adaptively restrain the varied band noise. Aggarwal and Majumdar [15] proposed the SSTV model by incorporating spatial correlation and spectral correlation into total variation. However, the SSTV model performs poorly in maintaining the complex spatial structure under high noise intensity. To overcome this limitation, Takemoto et al. [16] developed a graph-SSTV (GSSTV) model by introducing a weighted spatial difference operator, which reflects the spatial structure of HSI with a graph. Regrettably, most models are mainly designed to remove one or two types of noise, which leads to less competitive performance in handling mixed noise. More recently, the deep learning-based methods have achieved excellent performance due to their strong ability to learn prior knowledge, including DDS2M [18], SMDS-Net [19], and RCILD [20]. Nevertheless, these approaches are constrained by their reliance on extensive training datasets, which limits their generalizability to diverse real-world HSIs.

In practical applications, HSIs are often corrupted by multiple noises simultaneously, including Gaussian noise, impulse noise, deadlines, and stripes. Unfortunately, removing mixed noise is a challenging task. Recently, researchers have found that HSIs have low-rank properties in both the spatial and the spectral domains. Taking advantage of this property, a large number of low-rank regularization models [21], [22], [23] have been proposed for the task of removing mixed noise in HSIs and have shown outstanding performance. Among them, the most representative models include LRMR [24], NFF [25], SNLRSF [26], LRTF-DFR [27], and 3DCTV [28]. Zhang et al. [24] developed the LRMR model by decomposing the degraded HSI into low-rank and sparse components, which demonstrated excellent performance in removing mixed noise. However, the proposed method requires the rank and cardinality information to be specified in advance, which is a challenging task in real HSIs polluted by severe noise. To this end, Liu et al. [25] proposed the NFF model to automatically capture the intrinsic rank information. Cao et al. [26] proposed the SNLRSF model by utilizing subspace representation and nonlocal low-rank factorization, which has a good ability to retain local details. Unfortunately, the SNLRSF model cannot fully take advantage of the common characteristics of different frequency bands and ignores the local continuity of the spectrum. To solve this problem, Zheng et al. [27] developed the LRTF-DFR model by linking factor-based regularization to HSI priors, which enhanced the characterization capabilities of group sparsity and spectral continuity.

To sum up, existing models have achieved significant success in HSI denoising tasks. However, a careful examination of these models and methods reveals that they still have some limitations, which can be summarized as follows.

1) They lack a unified term to characterize the intrinsic structure of HSI. To address various specific issues, different regularization terms have been adopted. However, each regularization term comes with its own challenges. For instance, the nuclear norm tends to overpenalize larger singular values, resulting in suboptimal recovery of image texture details. Thus, to flexibly adopt different regularization terms for various problems, it is essential to establish a unified framework that facilitates a plug-and-play approach.
2) A multitude of models are limited in handling real HSI as they can only address one or two types of noise. Consequently, it is imperative to develop a model capable of removing mixed noise.
3) Given that low-rank components inherently possess sparse attribute information [29], [30], [31], where most of this sparse information constitutes image details, but many models overlook this sparsity. Accordingly, it is crucial to integrate sparse priors into the low-rank terms of the model.

To address these issues, this article proposes a novel generalized nonconvex low-rank framework, termed GNRSR, for HSI denoising tasks, which innovatively integrates a generalized nonconvex surrogate and a sparse regularizer. Our GNRSR shows excellent performance in preserving image details while effectively removing mixed noise, which boasts three key advantages. First, the generalized nonconvex surrogate supports plug-and-play nonconvex penalties that can be tailored to specific tasks. Second, the unified framework incorporates a universal optimizer, avoiding case-by-case nonconvex solvers. Third, the sparse regularizer preserves fine details, reducing oversmoothing in GNRSR. Compared with existing HSI denoising models, the main contributions and novelties of this article are summarized as follows.

1) A GNRSR framework with a unified nonconvex surrogate function and a sparse regularizer is proposed for the HSI denoising task, which fully exploits the low-rank and sparse prior knowledge in HSI. Therefore, the proposed GNRSR offers sufficient flexibility to handle mixed noise and enables a more accurate estimation of the original clean HSI.
2) A unified and efficient algorithmic framework based on inexact augmented Lagrange multiplier (ALM) is developed to address the formulated GNRSR model, where each subproblem can be addressed using proximal operators with closed-form solutions. The proposed algorithmic framework encompasses most existing methods as special cases.
3) Experimental results on two simulated and two real HSI datasets illustrate that our model outperforms other state-of-the-art models in both quantitative and visual performance.

The rest of this article is arranged as follows. Section II briefly reviews the related work. Section III describes the proposed GNRSR model and the corresponding algorithmic framework. Section IV provides the experimental performance of the proposed GNRSR on real HSI datasets. Finally, conclusions are drawn in Section V.

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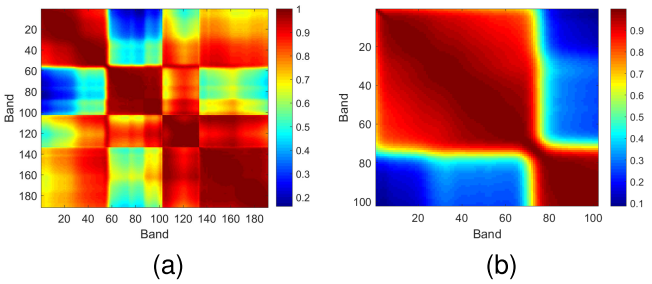


Fig. 1. Correlation coefficients between spectral bands in two HSI datasets. (a) HYDICE Washington dc Mall dataset. (b) Pavia Center dataset.

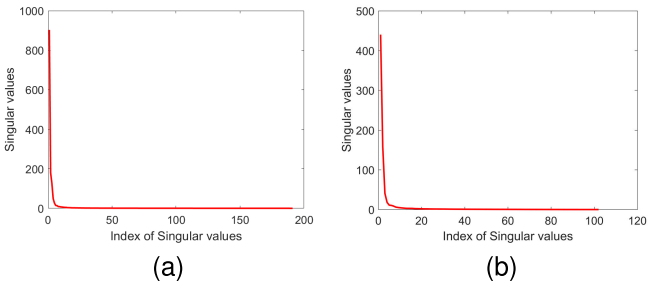


Fig. 2. Singular values of the Casorati matrix in two HSI datasets. (a) HYDICE Washington dc Mall dataset. (b) Pavia Center dataset.

II. Background

In this section, a brief review of the background related to HSI denoising tasks is provided.

A. Observation Model

In reality, due to the limitations of sensors and natural environments, the captured HSI is usually degraded by a mixture of Gaussian noise, impulse noise, stripes, deadlines, and so on. Accordingly, a degraded HSI $\mathcal{Y} \in \mathbb{R}^{m\times n\times B}$ can be regarded as an additive model

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{1}$$

where $\mathcal{X}$ is the clean HSI, $\mathcal{S}$ represents the sparse noise, and $\mathcal{N}$ denotes the Gaussian noise. The structure of the degraded HSI $\mathcal{Y}$ is a cube, which can be viewed as consisting of multiple subcubes of size $q \times q \times B$ centered at the pixel $(u, v)(1 \le u \le m, 1 \le v \le n)$ By reshaping each band of a subcube into a 1-D column vector, a matrix $Y \in \mathbb{R}^{q^2\times B}$ can be obtained. Thus, the observation model (1) can be converted into the matrix form

$$Y = X + S + N. \tag{2}$$

B. Low-Rank Property of HSI

HSIs have low-rank properties both spatially and spectrally. In the spatial dimension, like natural images, HSIs are composed of repetitive patterns, symmetries, and geometric structures, which in turn exhibit low-rank characteristics. In the spectral dimension, the high correlation between spectral bands (shown in Fig. 1) indicates the low-rank property.

Furthermore, for an HSI with dimensions $m \times n \times B$, the matrix with the size of $m \times n$ from each spectral band is reshaped columnwise into a vector of length mn, and these B vectors are then arranged into a matrix of size $mn \times B$, maintaining their original spectral band sequence, which is referred to as the Casorati matrix. The singular value distribution of the Casorati matrix shown in Fig. 2 also demonstrates the approximate low-rank property of HSI in the spectral dimension. The low-rank property of HSI proves that low-rank representation is an effective method for denoising.

C. Low-Rank Representation for HSI Denoising

Low-rank representation is a powerful tool for depicting the intrinsic structure of HSI. Therefore, how to utilize low-rank representation to denoise HSI has become a research focus for many researchers.

HSI denoising is to recover the clean image X from the degraded Y.

The original HSI denoising problem based on low-rank representation can be expressed as follows:

$$\min_{X,S} \ \text{rank}\,(X) + \lambda\|S\|_0, \quad \text{s.t. } Y = X + S \tag{3}$$

where $\lambda > 0$ is a regularization parameter. Because solving the optimization problem in (3) is NP-hard, most HSI denoising work focus on developing different low-rank regularization terms to approximate the rank function.

1) *Convex Low-Rank Representation:* A simple and straightforward relaxation method for problem (3) can be achieved by replacing l_0-norm with l_1-norm and replacing the rank function with the nuclear norm, thereby transforming it into the following convex optimization problem:

$$\min_{X,S} \ \|X\|_* + \lambda\|S\|_1 \quad \text{s.t. } Y = X + S. \tag{4}$$

This model shows excellent performance in recovering HSI degraded only by sparse noise. However, it performs poorly on problems that contain other noise. To address this limitation, Zhou et al. [32] considered HSI degraded by both sparse noise and Gaussian noise, and developed an HSI denoising model for removing mixed noise, which can be described as follows:

$$\min_{X,S} \ \|X\|_* + \lambda\|S\|_1, \quad \text{s.t. } \|Y - X - S\|_F \le \delta \tag{5}$$

where δ is a constant that corresponds to the standard deviation of Gaussian noise.

Although these HSI denoising methods based on convex low-rank approximation have made significant progress, they all exhibit inherent limitations. For example, the utilization of nuclear norm penalties leads to biased solutions, and using the weighted sum of tensor nuclear norms requires the introduction of additional information.

2) *Nonconvex Low-Rank Representation:* More and more evidence and works [33], [34] indicate that nonconvex low-rank representation offers significant advantages over its convex counterparts, such as reduced computational costs and an enhanced proximity to the global optimal solution [35]. Compared with traditional models, nonconvex low-rank representation offers superior flexibility for modeling complex and mixed noise

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distributions [36], [37], thereby enabling more robust denoising in real-world HSI scenarios. Consequently, methods based on nonconvex low-rank representation have garnered considerable interest from the research community. Xie et al. [38] proposed the nonconvex regularized low-rank and sparse (NonRLRS) matrix decomposition method for HSI denoising. The model is expressed as follows:

$$\min_{X,S} \; g(\sigma(X)) + \lambda g(S), \quad \text{s.t. } \|Y - X - S\|_F \le \delta \tag{6}$$

where $g(\cdot)$ is defined as a nonconvex regularizer, which can be formulated as follows:

$$g(X) = \sum_{j=1}^{B} \sum_{i=1}^{q^2} f_{ij}(X) \tag{7}$$

where $f_{ij}(X)$ is defined as follows:

$$f_{ij}(X) = \begin{cases} \phi\left(\dfrac{|X_{ij}|}{\|X\|_\infty}\right), & \text{if } \|X\|_\infty \neq 0 \\ 0, & \text{if } \|X\|_\infty = 0 \end{cases} \tag{8}$$

and the scalar function ϕ takes the form

$$\phi(t) = \text{sign}(t)(1+\epsilon)\frac{|t|}{|t|+\epsilon}. \quad t \in \mathbb{R}. \tag{9}$$

The NonRLRS model does not require predefined rank and cardinality information and exhibits better mixed noise denoising performance. However, the model does not fully exploit the inherent characteristics of HSI and performs poorly when HSI is only degraded by Gaussian noise. Recently, in order to better approximate the rank function to explore the low-rank structure of HSI, Liu et al. [25] proposed the NFF model. This model substitutes the nuclear norm with a nonconvex fraction function, demonstrating superior denoising performance in cases of both single and mixed noise, which can be formulated as follows:

$$\min_{X,S} \; \mathcal{P}_a(X) + \lambda\|S\|_1, \quad \text{s.t. } \|Y - X - S\|_F \le \delta \tag{10}$$

where the nonconvex fraction function $\mathcal{P}_a$ can be formulated as follows:

$$\mathcal{P}_a(X) = \sum_{i=1}^{m} \frac{a\sigma_i(X)}{1 + a\sigma_i(X)} \tag{11}$$

where $\sigma_i(X)$ is the ith singular value of X. In addition to these two models, numerous other nonconvex low-rank representation-based models also demonstrate superior denoising performance, significantly surpassing most convex low-rank representation-based HSI denoising models. However, they overlook the sparse attribute information inherent in the low-rank component.

D. Sparsity of HSI in the Transform Domain

Natural signals and images inherently have sparse representations in various analytical transform domains [e.g., discrete Fourier transform (DFT), discrete cosine transform (DCT), and Haar transform (HT)], which represent important features through a small number of large coefficients, while insignificant information is represented by coefficients close to zero [39]. Similarly, HSI also demonstrates sparsity after forward transformation, which captures the local features and detailed information of the image [40]. Fig. 3 illustrates that HSI has significant sparsity after forward transformation on both the Washington dc Mall dataset and the Pavia Center dataset. Given that low-rank components are inherently sparse representing detailed features such as image texture, most models overlook this aspect. Therefore, it is essential to incorporate the sparse prior of HSI in the transform domain into the model.

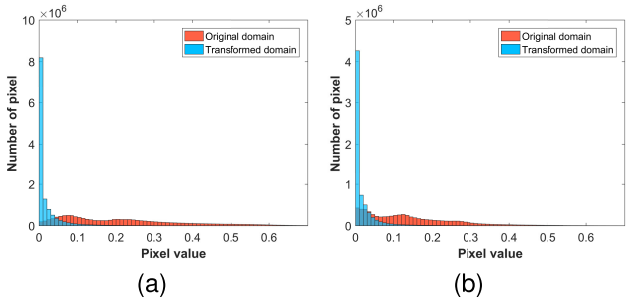


Fig. 3. Pixel value distribution in the original domain and the transformed domain in two HSI datasets. (a) HYDICE Washington dc Mall dataset. (b) Pavia Center dataset.

III. Proposed Model

In this section, we will propose a GNRSR framework for HSI denoising task. The key idea of the proposed GNRSR is that we establish a unified nonconvex surrogate function for matrix rank, rather than relying on the convex nuclear norm, while simultaneously introducing a sparse regularizer to characterize the sparse prior of HSI. We will propose a unified algorithmic framework for solving the proposed GNRSR and conduct a theoretical analysis of it.

Algorithm 1 Proposed Algorithm for HSI Denoising

1: **Input:** Degraded HSI $\mathcal{Y} \in \mathbb{R}^{m\times n\times B}$.
2: **Output:** Denoised HSI $\mathcal{X} \in \mathbb{R}^{m\times n\times B}$.
3: **Initialize:** s, q.
4: **for** each corrupted patch **do**
5: Convert corrupted cube patchto corrupted matrix patch$Y$.
6: X = GNRSR_model(Y).
7: Convert denoised matrixpatch X to denoisedcube patch.
8: **end for**
9: Aggregate all denoised cube patches and average the overlapping areas to get $\mathcal{X}$.

A. Problem Formulation

To better capture the inherent low-rank and sparse prior information in HSIs, we propose GNRSR model for HSI denoising tasks. The proposed GNRSR method has two parts: 1) patch grouping by exploiting the nonlocal self-similarity prior and the low-rank prior [41], [42] in the spectral dimension of HSI and 2) denoising the patch using nonconvex

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low-rank approximation by solving the GNRSR model. For a degraded HSI $\mathcal{Y} \in \mathbb{R}^{m\times n\times B}$, it is first divided into T corrupted patches with overlapping areas. These T patches have the same interval s, and the size of each is $q \times q \times B$. Therefore, the number of corrupted patches $T = \lceil (m-q)/s \rceil \times \lceil (n-q)/s \rceil$. According to Section II, the corruption patch $Y \in \mathbb{R}^{q^2\times B}$ in matrix form can be obtained. Then, these corrupted patches are denoised by the GNRSR model. The denoised patches X are reshaped back into $q\times q\times B$ cubes according to the inverse method and then replace the corrupted patches in the original position. After processing each corrupted patch, the denoised HSI is obtained by averaging the overlap between patches. The complete procedure of the proposed HSI denoising method is summarized in Algorithm 1.

On one hand, the proposed GNRSR method integrates numerous nonconvex surrogates into a unified nonconvex surrogate structure, enabling the application of different nonconvex penalty functions for specific denoising requirements. On the other hand, it effectively removes noise from damaged patches while preserving the texture details of the HSI. Mathematically, the proposed GNRSR model can be formulated as follows:

$$\begin{aligned} \min_{X,S}\ & \Phi(X) + \lambda\|S\|_1 + \gamma\|\mathcal{G}(X)\|_1 \\ \text{s.t.}\ & \|Y - X - S\|_F \le \delta \end{aligned} \tag{12}$$

where $\Phi(X) = \sum_{i=1}^{m} \phi(\sigma_i(X))$ is the generalized nonconvex surrogate with a wide range of nonconvex penalty functions, and $\mathcal{G}(\cdot)$ denotes a linear and unitary transform operator. It is worth noting that, according to the recommendations in [49], the nonconvex penalty function $\phi(\cdot)$ is required to be concave, nondecreasing, and differentiable on $\mathbb{R}_{\ge 0}$, with $\phi(0) = 0, \lim_{x\to+\infty} \phi(x)/x = 0$. Furthermore, its gradient ϕ' should be convex and continuous. Obviously, several popular nonconvex surrogates, such as MCP [44], ETP [46], ℓ_p-norm [47], and NFF [48] listed in Table I, have been successfully integrated into GNRSR. Furthermore, as can be seen from Fig. 4, the nonconvex surrogate functions can better approximate the rank function. Compared with the existing models, the proposed GNRSR model has the following two advantages.

1) *It Is Plug-and-Play and Can Flexibly Apply Various Nonconvex Surrogates for Specific Challenges:* By selecting scenario-specific penalties within its generalized nonconvex framework, the GNRSR model ensures superior denoising performance through optimal singular value recovery of HSIs across diverse noise conditions.
2) *It Can Recover Clean HSI From Degraded HSI With High Accuracy:* By embedding a transform-domain sparse regularizer that exploits the inherent sparsity of HSIs, the GNRSR model effectively preserves image details through further improving singular value recovery accuracy.

B. Optimization Method

To facilitate the subsequent processing, we introduce a variable W to represent the DCT operation on X, i.e., $W = \mathcal{G}(X)$. Thus, problem (12) can be recast as follows:

$$\begin{aligned} \min_{X,S}\ & \Phi(X) + \lambda\|S\|_1 + \gamma\|W\|_1 \\ \text{s.t.}\ & \|Y - X - S\|_F \le \delta \\ & W = \mathcal{G}(X). \end{aligned} \tag{13}$$

Then the augmented Lagrangian function for (13) can be written as follows:

$$\begin{aligned} \mathcal{L}(X,S,W,P,Q;\mu) = {} & \Phi(X) + \lambda\|S\|_1 + \gamma\|W\|_1 \\ & + \langle P, Y - X - S\rangle \\ & + \frac{\mu}{2}\|Y - X - S\|_F^2 \\ & + \langle Q, W - \mathcal{G}(X)\rangle \\ & + \frac{\mu}{2}\|W - \mathcal{G}(X)\|_F^2 \end{aligned} \tag{14}$$

where P and Q are the Lagrange multipliers, $\mu > 0$ is the penalty parameter, and $\langle\cdot,\cdot\rangle$ is the inner product operator. Since optimization problem (14) involves nonconvex low-rank regularizers, the resultant problem is nonconvex and even nonsmooth, making it much more challenging to solve directly. Besides, the presence of $\mathcal{G}(X)$ in problem (14) further increases the difficulty of solving it. To this end, we design an efficient and effective algorithm to solve such problem using the inexact ALM framework. Specifically, this optimization problem can be addressed by iteratively updating X, S, W, P, Q, and μ respectively via the following scheme:

$$\begin{cases} X_{k+1} = \arg\min_X \mathcal{L}(X, S_k, W_k, P_k, Q_k; \mu_k) & (15)\\ S_{k+1} = \arg\min_S \mathcal{L}(X_{k+1}, S, W_k, P_k, Q_k; \mu_k) & (16)\\ W_{k+1} = \arg\min_W \mathcal{L}(X_{k+1}, S_{k+1}, W, P_k, Q_k; \mu_k) & (17)\\ P_{k+1} = P_k + \mu_k(Y - X_{k+1} - S_{k+1}) & (18)\\ Q_{k+1} = Q_k + \mu_k(W_{k+1} - \mathcal{G}(X_{k+1})) & (19)\\ \mu_{k+1} = \min\{\rho\mu_k, \mu_{\max}\} & (20) \end{cases}$$

where $\rho > 1$ is used to reduce the number of iterations. The details of the updates for X, S, and W are shown below.

1) Update X_{k+1}. Through fixingthe other variables, X_{k+1} can beupdate by

$$\begin{aligned} X_{k+1} &= \arg\min_X \mathcal{L}(X, S_k, W_k, P_k, Q_k; \mu_k) \\ &= \arg\min_X \Phi(X) + \langle P, Y - X - S_k\rangle \\ &\quad + \frac{\mu_k}{2}\|Y - X - S_k\|_F^2 + \langle Q, W_k - \mathcal{G}(X)\rangle \\ &\quad + \frac{\mu_k}{2}\|W_k - \mathcal{G}(X)\|_F^2 \\ &= \arg\min_X \Phi(X) + \frac{\mu_k}{2}\left\|X - \left(Y - S_k + \frac{P_k}{\mu_k}\right)\right\|_F^2 \\ &\quad + \frac{\mu_k}{2}\left\|\mathcal{G}(X) - \left(W_k + \frac{Q_k}{\mu_k}\right)\right\|_F^2. \end{aligned} \tag{21}$$

Note that the above problem (21) is challenging to solve directly because of the DCT operator $\mathcal{G}(\cdot)$. Fortunately, Merhav and Kresch [50] have shown that DCT is a unitary transform, which can be energy conservation, i.e., $\|A\|_F^2 = \|B\|_F^2$ with $A = \mathcal{F}(B)$, where $\mathcal{F}(\cdot)$ is its

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inverse (i.e., adjoint) operator. Thus, the problem (21) can be rewritten as follows:

$$\begin{aligned} X_{k+1} &= \arg\min_X \Phi(X) + \frac{\mu_k}{2}\left\|X - \left(Y - S_k + \frac{P_k}{\mu_k}\right)\right\|_F^2 \\ &\quad + \frac{\mu_k}{2}\left\|\mathcal{F}\left(\mathcal{G}(X) - \left(W_k + \frac{Q_k}{\mu_k}\right)\right)\right\|_F^2 \\ &= \arg\min_X \Phi(X) + \frac{\mu_k}{2}\left\|X - \left(Y - S_k + \frac{P_k}{\mu_k}\right)\right\|_F^2 \\ &\quad + \frac{\mu_k}{2}\left\|X - \mathcal{F}\left(W_k + \frac{Q_k}{\mu_k}\right)\right\|_F^2 \\ &= \arg\min_X \sum_{i=1}^{r} \phi(\sigma_i(X)) + \mu_k \left\|X - \frac{1}{2}(Y - S_k \right. \\ &\quad \left. + \frac{P_k}{\mu_k} + \mathcal{F}\left(W_k + \frac{Q_k}{\mu_k}\right)\right)\right\|_F^2. \end{aligned} \tag{22}$$

Then the above optimization problem (22) can be solved by following Proposition 1.

Proposition 1 (Generalized Singular Value Thresholding (GSVT) [49]): Let $U\mathrm{Diag}(\sigma(Q))V^\top$ be the SVD of Q, the closed-form solution to the following problem

$$\arg\min_X \sum_{i=1}^{r} \phi(\sigma_i(X)) + \frac{\eta}{2}\|X - Q\|_F^2 \tag{23}$$

takes the form

$$X^* = U\mathrm{Diag}(\varrho^*)V^\top \tag{24}$$

where

$$\varrho_i^* = \arg\min_{\varrho_i \geq 0} \phi(\varrho_i) + \frac{\eta}{2}(\varrho_i - \sigma_i(Q))^2. \tag{25}$$

According to Proposition 1, by replacing $\phi(x)$ in (22) with a specific nonconvex function, we can obtain the optimal solution of problem (22) as follows:

$$X_{k+1} = \hat{U}\mathrm{Diag}(\varrho^*)\hat{V}^\top \tag{26}$$

where $R_k = \hat{U}\mathrm{Diag}(\sigma(R_k))\hat{V}^\top$ is the SVD of the matrix $R_k = (1/2)(Y - S_k + P_k/\mu_k + \mathcal{F}(W_k + Q_k/\mu_k))$. Update S_{k+1}. To update S_{k+1}, we need to consider the following subproblem:

$$\begin{aligned} S_{k+1} &= \arg\min_S \mathcal{L}(X_{k+1}, S, W_k, P_k, Q_k; \mu_k) \\ &= \arg\min_S \lambda\|S\|_1 + \langle P, Y - X_{k+1} - S\rangle \\ &\quad + \frac{\mu_k}{2}\|Y - X_{k+1} - S\|_F^2 \\ &= \arg\min_S \frac{\lambda}{\mu_k}\|S\|_1 \\ &\quad + \frac{1}{2}\left\|S - \left(Y - X_{k+1} + \frac{P_k}{\mu_k}\right)\right\|_F^2. \end{aligned}$$

The above problem (27) is a standard ℓ_1-norm regularization optimization problem. To obtain the closed-form solution, we employ the common methodology presented in [51]. For a given matrix $B \in \mathbb{R}^{m\times n}$, the optimal solution to the problem

$$\min_S \tau\|S\|_1 + \frac{1}{2}\|S - B\|_F^2 \tag{27}$$

is $S^* = \mathcal{S}_\tau(x)$, where x represents an element of the matrix B, and $\mathcal{S}_\tau(\cdot)$ is the soft-thresholding operator defined as $\mathcal{S}_\tau(x) = \mathrm{sign}(x)\max(|x| - \tau, 0)$. Therefore, the closed-form solution of optimization problem (27) can be expressed as follows:

$$S_{k+1} = \mathcal{S}_{\frac{\lambda}{\mu_k}}\left(Y - X_{k+1} + \frac{P_k}{\mu_k}\right). \tag{28}$$

Update W_{k+1}. To obtain W_{k+1}, the following subproblem needs to be considered:

$$\begin{aligned} W_{k+1} &= \arg\min_W \mathcal{L}(X_{k+1}, S_{k+1}, W, P_k, Q_k; \mu_k) \\ &= \arg\min_W \gamma\|W\|_1 + \langle Q, W - \mathcal{G}(X_{k+1})\rangle \\ &\quad + \frac{\mu_k}{2}\|W - \mathcal{G}(X_{k+1})\|_F^2 \\ &= \arg\min_W \frac{\gamma}{\mu_k}\|W\|_1 \\ &\quad + \frac{1}{2}\left\|W - \left(\mathcal{G}(X_{k+1}) - \frac{Q_k}{\mu_k}\right)\right\|_F^2. \end{aligned} \tag{29}$$

Similar to the update of S, the closed-form solution to problem (29) is as follows:

$$W_{k+1} = \mathcal{S}_{\frac{\gamma}{\mu_k}}\left(\mathcal{G}(X_{k+1}) - \frac{Q_k}{\mu_k}\right). \tag{30}$$

The above update process (15)–(20) is continuously iterated until the stopping criterion is met and the algorithm terminates. The stopping conditions are defined as follows:

$$\max\left\{\begin{matrix}\|Y - X_{k+1} - S_{k+1}\|_F/\|Y\|_F \\ \|X_{k+1} - X_k\|_F\end{matrix}\right\} \leq \epsilon \tag{31}$$

where ϵ is a small tolerance.

Algorithm 2 Solving Problem (12) via Inexact ALM

Input: Corrupted matrix patch $Y \in \mathbb{R}^{q^2\times B}$.
Output: Denoised matrix patch $X \in \mathbb{R}^{q^2\times B}$.
Initialize: $S_0 = P_0 = Q_0 = 0$, λ, γ, μ_0, ϵ, $k = 0$, K.
while *Stopping criterion is not satisfied* **do**
  Update X_{k+1} by (26).
  Update S_{k+1} by (28).
  Update W_{k+1} by (30).
  Update P_{k+1} by (18).
  Update Q_{k+1} by (19).
  Update μ_{k+1} by (20).
  $k \leftarrow k + 1$.
end while

Based on the above analysis, an iterative scheme approach for the problem (12) is derived, which is summarized in Algorithm 2.

C. Algorithm Analysis

In this section, we first present the following theorem to establish the convergence properties of Algorithm 2 under mild conditions. This result provides a solid theoretical guarantee for the practical application of our method. The detailed proof is provided in the supplementary material.

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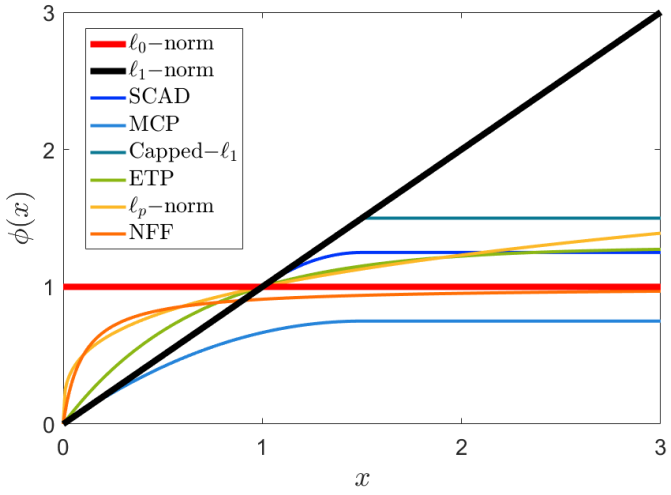


Fig. 4. Approximation of rank function with different nonconvex penalty function.

Theorem 1: Let $\{(X_k, S_k, W_k, P_k, Q_k)\}$ be the sequence generated by Algorithm 2, then we have the following conditions.

1) The sequence $\{(X_k, S_k, W_k, P_k, Q_k)\}$ is bounded.
2) Any accumulation point of the sequence $\{(X_k, S_k, W_k, P_k, Q_k)\}$ is a KKT point.

To further demonstrate the convergence, Fig. 5 shows the variation of relative errors $\|Y - X_{k+1} - S_{k+1}\|_F$ and $\|X_{k+1} - X_k\|_F$ in Cases 1–6 on the two datasets. After several iterations, the relative error has stabilized across all cases. Consequently, the proposed Algorithm 2 effectively addresses the issues arising from the GNRSR model in (12).

Then the time complexity of Algorithm 2 is introduced in detail. Assume that forward transform operator $\mathcal{G}(\cdot)$ used in Algorithm 2 is the DCT operator. The main cost of Algorithm 2 contains three parts. The first is updating X. Its main computations come from the inverse transform of DCT and the SVD operation, which respectively take $\mathcal{O}(q^2 B \log q^4 B^2)$ and $\mathcal{O}(q^2 B^2)$, and the rest costs $\mathcal{O}(B)$. The second part is updating S. Its computational complexity is $\mathcal{O}(q^2 B)$. The third part is updating W with a computational complexity of $\mathcal{O}(q^2 B \log q^4 B^2 + q^2 B)$, where $\mathcal{O}(q^2 B \log q^4 B^2)$ is the computational complexity of DCT. The computational complexity of the remaining parts is constant time. The dominant cost of Algorithm 2 is to update X, and its time complexity is $\mathcal{O}(q^2 B \log q^4 B^2 + q^2 B^2)$. The main cost of Algorithm 1 is incurred in step 6, which is actually Algorithm 2. Consequently, the total computational complexity of the GNRSR method is $\mathcal{O}(\mathrm{TK}(q^2 B \log q^4 B^2 + q^2 B^2))$, where K is the maximum number of iterations.

IV. Experimental Results and Discussion

In this section, we conduct experiments with simulated and real datasets to qualitatively and quantitatively demonstrate the efficiency and practicality of the proposed GNRSR. We compare our method with nine state-of-the-art approaches, including two traditional methods (LRMR [24] and NonLRMA [52]), two deep learning-based methods (SMDS [19] and RCILD [20]), and five newly proposed methods (WNLRATV [53], NFF [25], FastHyMix [54], 3DCTV [28], and FallHyDe [55]). All the parameters involved in comparison algorithms are fine-tuned or automatically selected as described in their reference articles. SMDS and RCILD are performed on Ubuntu 20.04.6 with an NVIDIA T4 GPU, and all remaining experiments are performed on Windows 11 with the 11th Intel Core i7-11800H 2.30-GHz CPU, 24 GB memory. Specifically, to better demonstrate the excellent denoising performance of our algorithm, we set the forward transform operator $\mathcal{G}(\cdot)$ to the DCT operator and adopt the NFF and ℓ_p-norm as nonconvex low-rank surrogates. This choice is motivated by the DCT's capability to achieve high energy compaction and smooth reconstructions without boundary artifacts.

A. Simulated Experiments

Two widely used public hyperspectral datasets, namely the Washington dc Mall dataset and the Pavia Center dataset, are selected to verify the performance of our proposed method. Specifically, subimages of size $256 \times 256 \times 191$ in the WDC dataset and $256 \times 256 \times 102$ in the Pavia Center dataset are used in the experiments, which are shown in Fig. 6(a) and (b), respectively. In addition, as in [24], each band of the dataset is normalized to the range [0, 1].

In the simulated experiments, we will consider the following six scenarios.

Case 1 (Gaussian Noise): All bands are corrupted by Gaussian noise with zero mean and variance randomly selected from [0.05, 0.15].

Case 2 (Gaussian Noise): All bands are corrupted by Gaussian noise with zero mean and variance randomly selected from [0.15, 0.25].

Case 3 (Gaussian Noise): All bands are corrupted by Gaussian noise with zero mean and variance randomly selected from [0.25, 0.35].

Case 4 (Gaussian Noise + Impulse noise): Based on Case 1, impulse noise with intensity sampled in [0.05, 0.15] is added to all bands.

Case 5 (Gaussian Noise + Stripes + Deadlines): Based on Case 1, stripes with the width in [3, 5] and deadlines with the width in [1, 3] are added to the selected 40% of the bands.

Case 6 (Mixed Noise): All the above types of noise are added to the original HSI. Specifically, all the bands are corrupted by Gaussian noise, impulse noise, stripes, and deadlines as mentioned in Cases 1, 4, and 5.

To quantitatively evaluate the experimental results, we selected four widely used metrics, which are peak signal-to-noise ratio (PSNR), Structural Similarity Index Measurement (SSIM), Feature Similarity Index Measurement (FSIM), and Erreur Relative Globale Adimensionnelle de Synthese (ERGAS). In particular, the higher the PSNR, SSIM, and FSIM values, the better the denoising effect, and the lower the ERGAS value, the better the denoising results.

1) Experimental Results on WDC Dataset: Table II lists the quantitative evaluation results of different methods on the WDC dataset under different noise cases. The best result among all methods is marked in bold, and the second best result is underlined. It can be observed that our methods outperform the comparative methods in almost all of the four

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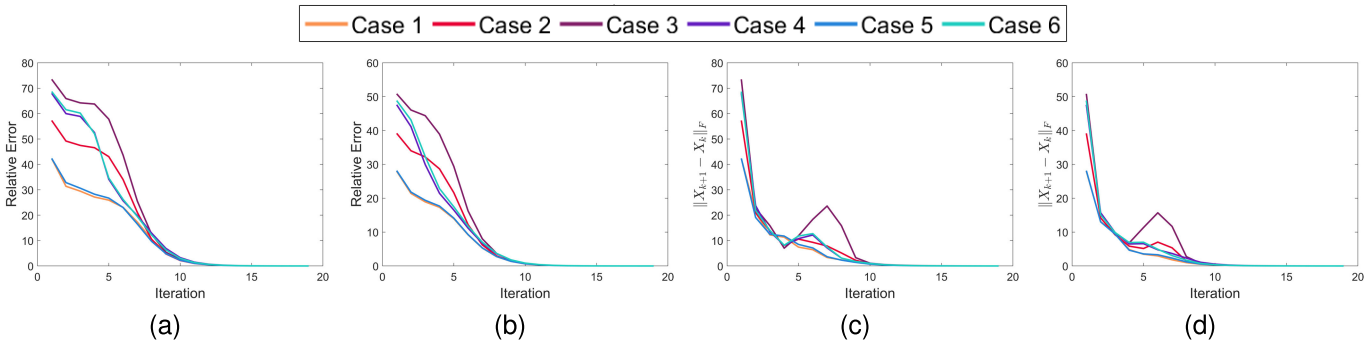


Fig. 5. Relative error variation and clean HSI variation of the GNRSR solver with the number of iterations for simulation cases 1–6. (a) Relative error variation on HYDICE Washington dc Mall dataset. (b) Relative error variation on Pavia Center dataset. (c) $\|X_{k+1} - X_k\|_F$ on HYDICE Washington dc Mall dataset. (d) $\|X_{k+1} - X_k\|_F$ on Pavia Center dataset.

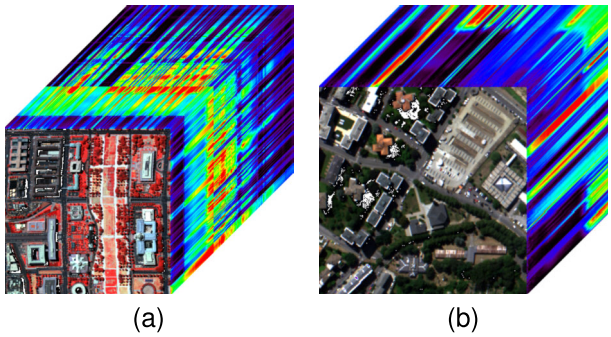


Fig. 6. (a) HYDICE Washington dc Mall dataset. (b) Pavia Center dataset.

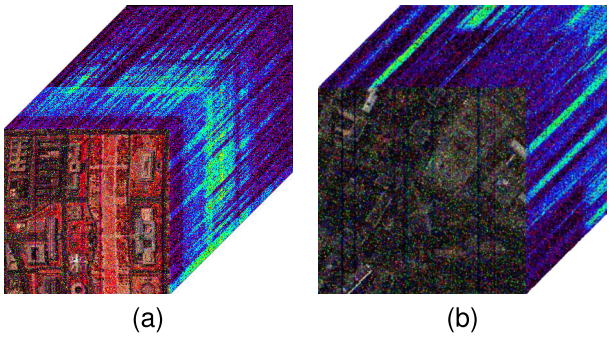


Fig. 7. (a) Corrupted HYDICE Washington dc Mall dataset under Case 6. (b) Corrupted Pavia Center dataset under Case 6.

TABLE I
SOME POPULAR NONCONVEX LOW-RANK REGULARIZERS

Name	Penalty Function $\phi(x)$
SCAD [43]	$\begin{cases} \beta\|x\|, & \|x\| \le \beta, \\ \dfrac{2\beta\alpha\|x\| - x^2 - \beta^2}{2(\alpha-1)}, & \beta < \|x\| < \alpha\beta, \\ \dfrac{\beta^2(\alpha+1)}{2}, & \|x\| > \alpha\beta. \end{cases}$
MCP [44]	$\begin{cases} \beta\|x\| - \dfrac{x^2}{2\alpha}, & \|x\| \le \alpha\beta, \\ \dfrac{1}{2}\alpha\beta^2, & \|x\| > \alpha\beta. \end{cases}$
Capped-ℓ_1 [45]	$\begin{cases} \beta\|x\|, & \|x\| \le \alpha, \\ \beta\alpha, & \|x\| > \alpha. \end{cases}$
ETP [46]	$\dfrac{\beta}{1-e^{-\alpha}}(1-e^{-\alpha x}).$
ℓ_p-norm [47]	$\beta\|x\|^p.$
NFF [48]	$\dfrac{\beta\alpha x}{1+\alpha x}.$

metrics across six scenarios. In the case of low-intensity Gaussian noise (Case 1), the state-of-the-art FastHyMix method and RCILD method perform best in the FSIM metric but are inferior to the proposed GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$ methods in the other three metrics. More importantly, it is evident that as the noise intensity increases (from Cases 1 to 3), the superiority of the proposed methods becomes more pronounced.

To provide a more comprehensive comparison of the denoising capabilities of each method in terms of visual quality, we select the HSI from the WDC dataset under the most complex mixed noise (Case 6) as the denoising target [shown in Fig. 7(a)]. Fig. 8 presents the denoising results of the pseudo-color image consisting of bands 69, 106, and 151 in the WDC Mall dataset. Fig. 9 shows the denoising results in band 106. It can be clearly observed that the denoising results of LRMR and FallHyDe obviously retain some Gaussian noise and impulse noise. NonLRMA and FastHyMix cause significant spectral distortion. RCILD cannot remove deadlines and impulse noise. SMDS excessive denoising causes textures in the image to be too smooth. WNLRATV, NFF, and 3DCTV yield comparable results in terms of mixed noise removal. However, in the upper building in the enlarged area marked with a red box, NFF cannot restore the spectral information well, resulting in slight color differences in some areas compared to the original image; WNLRTV smooths some details, such as black spots on the building; and 3DCTV leaves a small amount of residual deadlines that cannot be completely eliminated. Compared with other methods, the proposed GNRSR$_{\text{NFF}}$ achieved superior visual quality by effectively eliminating all mixed noise and preserving fine details.

To more intuitively demonstrate the effectiveness of different methods in recovering spectral information, Fig. 10 shows the spectral difference between the clean image and the restored image at pixel (38, 60) in Case 6 in the WDC dataset. From Fig. 10(a), it can be observed that the complex mixed noise seriously destroyed the spectral features, resulting in drastic fluctuations in the curve. LRMR, NonLRMA, SMDS, RCILD, FastHyMix, and FallHyDe cannot restore

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TABLE II
QUANTITATIVE RESULTS OF COMPARISON METHODS UNDER DIFFERENT CASES FOR WDC DATASET

Cases	Indexes	Noised	LRMR	NonLRMA	SMDS	RCILD	WNLRATV	NFF	FastHyMix	3DCTV	FallHyDe	GNRSR$_{\text{NFF}}$	GNRSR$_{\ell_p}$
Case 1	**PSNR** ↑	21.157	32.897	29.686	32.403	35.973	34.349	34.937	35.923	35.706	34.557	**36.680**	36.647
	SSIM ↑	0.462	0.926	0.901	0.923	0.960	0.941	0.948	0.961	0.958	0.943	0.963	**0.964**
	FSIM ↑	0.755	0.963	0.942	0.962	**0.982**	0.966	0.969	**0.982**	0.975	0.976	0.980	0.980
	ERGAS ↓	341.475	88.679	125.106	92.621	72.727	74.120	67.742	65.314	64.412	80.983	**55.173**	55.401
Case 2	**PSNR** ↑	15.491	26.620	24.102	27.998	28.829	28.479	30.070	28.819	31.108	27.593	**32.195**	31.837
	SSIM ↑	0.212	0.812	0.745	0.860	0.882	0.823	0.863	0.883	0.886	0.831	**0.912**	0.906
	FSIM ↑	0.600	0.918	0.870	0.943	**0.955**	0.894	0.930	**0.955**	0.937	0.937	0.954	0.953
	ERGAS ↓	635.046	185.292	238.107	164.568	172.522	147.959	116.372	154.129	107.087	185.662	**92.517**	96.520
Case 3	**PSNR** ↑	12.451	22.719	21.962	23.844	24.148	25.274	27.263	24.151	28.558	23.336	**29.350**	29.081
	SSIM ↑	0.119	0.695	0.683	0.778	0.795	0.712	0.782	0.798	0.810	0.737	**0.855**	0.852
	FSIM ↑	0.512	0.877	0.858	0.915	0.928	0.820	0.891	0.927	0.901	0.908	**0.929**	0.928
	ERGAS ↓	900.034	294.890	316.492	270.933	297.091	209.572	160.314	264.361	142.213	305.438	**129.340**	133.548
Case 4	**PSNR** ↑	13.732	25.125	23.388	26.565	28.033	34.290	34.318	26.938	34.905	25.754	**36.019**	35.969
	SSIM ↑	0.204	0.765	0.726	0.844	0.874	0.939	0.941	0.856	0.950	0.763	0.958	**0.959**
	FSIM ↑	0.601	0.901	0.865	0.938	0.953	0.963	0.966	0.914	0.970	0.915	0.977	**0.978**
	ERGAS ↓	799.592	221.533	261.383	191.503	190.773	74.663	72.454	183.032	70.113	228.054	**59.405**	59.716
Case 5	**PSNR** ↑	20.514	31.486	29.100	31.429	34.002	33.849	33.492	33.973	34.877	33.514	**35.335**	34.951
	SSIM ↑	0.441	0.913	0.889	0.913	0.944	0.936	0.939	0.946	0.951	0.935	**0.958**	0.956
	FSIM ↑	0.741	0.954	0.934	0.955	0.969	0.962	0.963	0.972	0.971	0.970	**0.976**	0.975
	ERGAS ↓	363.919	104.659	133.846	104.525	92.208	77.560	80.706	82.013	71.594	88.608	**66.156**	69.621
Case 6	**PSNR** ↑	13.608	25.074	23.437	26.555	28.144	33.781	32.986	27.058	34.271	25.827	**34.743**	34.340
	SSIM ↑	0.197	0.757	0.721	0.839	0.870	0.934	0.933	0.856	0.944	0.759	**0.953**	0.952
	FSIM ↑	0.593	0.895	0.861	0.933	0.948	0.960	0.960	0.914	0.968	0.910	**0.973**	0.972
	ERGAS ↓	811.031	223.278	260.449	192.820	187.986	79.600	85.803	180.235	76.444	227.508	**70.902**	74.614
Time	**Average**	-	296.4	1055.8	1247.2	130.8	861.9	189.8	30.6	265.8	**2.2**	264.1	324.8

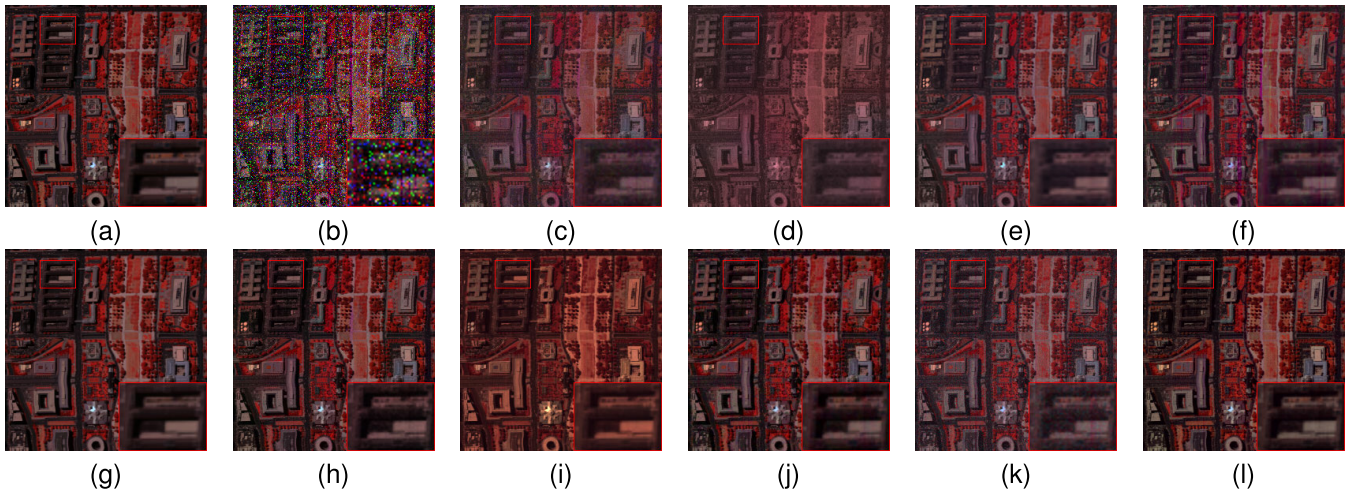


Fig. 8. Denoising results of the WDC Mall. (a) False-color original image with bands (R: 69, G: 106, and B: 151). (b) Noisy image. (c) LRMR. (d) NonLRMA. (e) SMDS. (f) RCILD. (g) WNLRATV. (h) NFF. (i) FastHyMix. (j) 3DCTV. (k) FallHyDe. (l) GNRSR$_{\text{NFF}}$.

the original spectral curve. WNLRATV, NFF, 3DCTV, and GNRSR demonstrate superior spectrum recovery performance compared to other methods.

2) Experimental Results on Pavia Center Dataset: The quantitative evaluation values of each method on Pavia Center dataset are presented in Table III. The superiority of our approaches over various other methods can be seen in Table III, as it consistently achieves the highest performance in the majority of different cases. In the case of complex mixed noise (Case 6), the denoising performance of the proposed GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$ methods are superior to the state-of-the-art WNLRATV, FastHyMix, and 3DCTV methods. In addition, Fig. 11 shows the PSNR and SSIM values for each band in the Pavia Center dataset under Cases 3 and 6. Clearly, the proposed methods demonstrate that they effectively remove high-intensity Gaussian noise and complex mixed noise. It also showcases their superiority in preserving the spatial structure of the original HSI and achieving an accurate low-rank approximation.

The degraded HSI of the Pavia Center dataset under Case 6 is shown in Fig. 7(b). Fig. 12 displays the denoising results of the pseudo-color image composed of bands 55, 35, and 15 in the Pavia Center dataset. Fig. 13 shows the denoising results in band 15. As can be seen, LRMR can only remove part of Gaussian noise and impulse noise. NonLRMA, SMDS, and FallHyDe induce substantial spectral distortion. A small number of deadlines still remain in the results of RCILD, FastHyMix, and 3DCTV. NFF and WNLRATV eliminate almost all noise, but smooth out some image details, such as white lines on buildings in the magnified area. GNRSR$_{\text{NFF}}$ achieves the best visual results, which preserve image details while removing noise.

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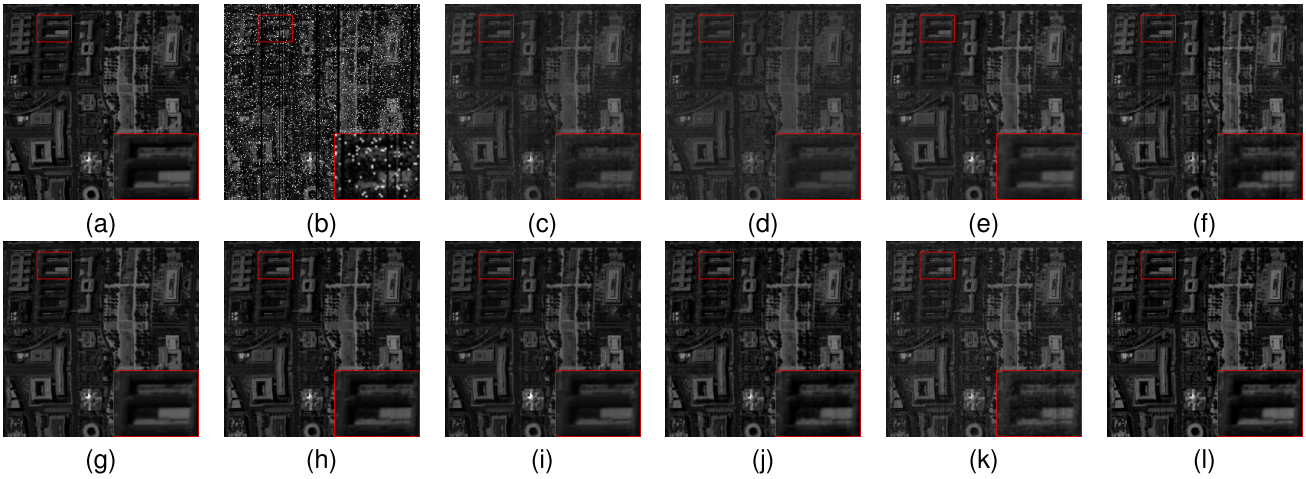


Fig. 9. Denoising results of the WDC Mall. (a) Original image with band 106. (b) Noisy image. (c) LRMR. (d) NonLRMA. (e) SMDS. (f) RCILD. (g) WNLRATV. (h) NFF. (i) FastHyMix. (j) 3DCTV. (k) FallHyDe. (l) GNRSR$_{\text{NFF}}$.

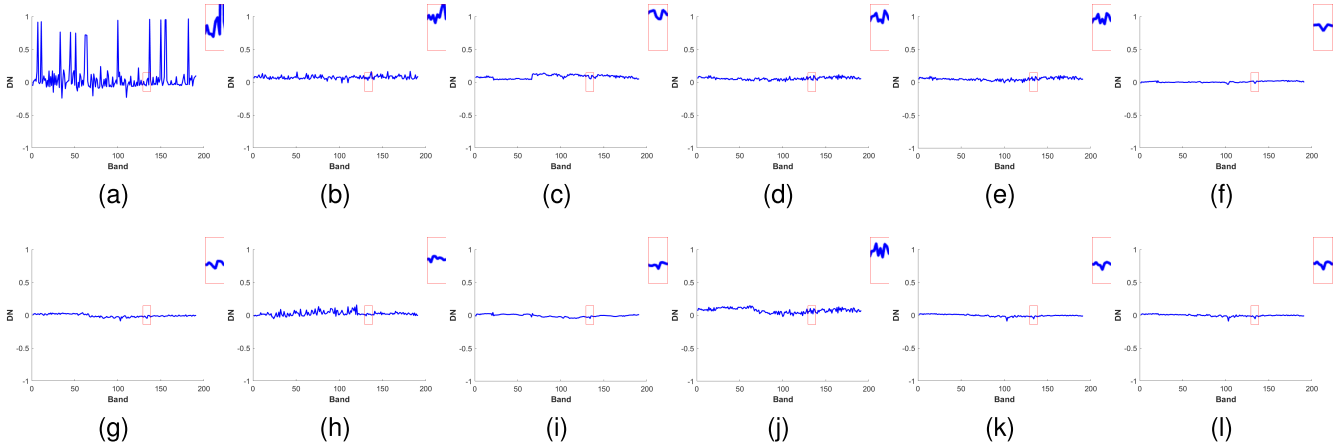


Fig. 10. Difference between the spectrum of the noise-free pixel (38, 60) and the corresponding restored pixels by different methods in the Washington dc Mall image in Case 6. (a) Noisy. (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

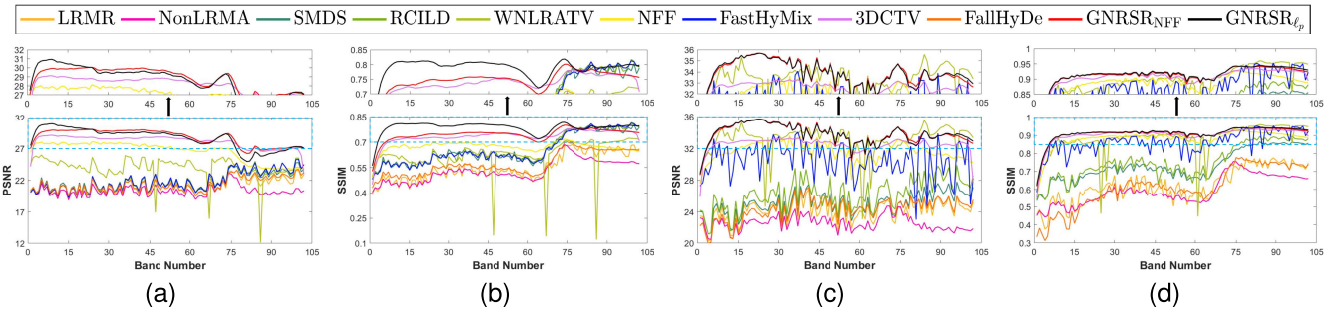


Fig. 11. PSNR and SSIM values of each band in the Pavia Center dataset in Cases 3 and 6. (a) PSNR value of Case 3. (b) SSIM value of Case 3. (c) PSNR value of Case 6. (d) SSIM value of Case 6.

Fig. 14 illustrates the spectral difference between the clean image and the restored image at pixel (17, 103) in Case 6 of the Pavia Center dataset. The rapid fluctuations of the curves in Fig. 14(a) indicate that most bands are severely polluted. LRMR, NonLRMA, SMDS, RCILD, NFF, FastHyMix, and FallHyDe cannot restore the original spectral curve. WNLRATV is able to recover most of the spectrum, but some bands still have residual sharp noise. 3DCTV and GNRSR achieve the best spectral restoration results and maintain spectral reflectance characteristics well.

B. Real Experiments

We selected two HSI datasets, namely HYDICE Urban and AVIRIS Indian Pines, to further evaluate the performance of GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$ on real-world data. As in the simulated data experiments, the pixels of the HSI are also normalized to [0, 1] for subsequent processing.

1) Experimental Results on AVIRIS Indian Pines Dataset: The AVIRIS Indian Pines dataset, acquired by the NASA AVIRIS sensor, is widely employed and recognized in the hyperspectral research community, with a size of $145 \times 145 \times$

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TABLE III
QUANTITATIVE RESULTS OF COMPARISON METHODS UNDER DIFFERENCE CASES FOR PAVIA CENTER DATASET

Cases	Indexes	Noised	LRMR	NonLRMA	SMDS	RCILD	WNLRATV	NFF	FastHyMix	3DCTV	FallHyDe	GNRSR$_{\text{NFF}}$	GNRSR$_{\ell_p}$
Case 1	**PSNR** ↑	21.592	31.455	29.067	31.159	33.439	34.085	33.378	33.624	33.496	32.538	35.195	**35.273**
	SSIM ↑	0.386	0.837	0.827	0.861	0.887	0.893	0.903	0.891	0.917	0.866	0.919	**0.921**
	FSIM ↑	0.713	0.950	0.933	0.953	**0.973**	0.962	0.956	**0.973**	0.961	0.966	0.967	0.967
	ERGAS ↓	384.946	122.733	158.323	126.984	107.362	90.620	96.699	98.975	97.368	113.495	78.228	**77.516**
Case 2	**PSNR** ↑	15.941	25.149	23.149	25.948	26.469	28.480	29.036	26.442	30.195	25.684	31.009	**31.066**
	SSIM ↑	0.164	0.666	0.629	0.757	0.768	0.764	0.784	0.768	0.823	0.687	0.835	**0.845**
	FSIM ↑	0.544	0.891	0.849	0.928	**0.938**	0.895	0.906	0.935	0.916	0.910	0.928	0.928
	ERGAS ↓	715.609	252.452	312.124	233.663	233.003	169.635	156.770	223.296	139.298	248.711	125.215	**124.227**
Case 3	**PSNR** ↑	12.777	21.041	20.421	21.782	21.992	23.879	26.616	21.965	28.110	21.456	**28.867**	28.653
	SSIM ↑	0.087	0.516	0.534	0.656	0.663	0.622	0.675	0.660	0.736	0.575	0.750	**0.788**
	FSIM ↑	0.452	0.833	0.825	0.898	**0.906**	0.809	0.862	0.893	0.875	0.874	0.893	0.894
	ERGAS ↓	1025.321	402.766	428.547	378.533	382.869	291.287	206.372	368.884	175.781	397.655	**159.528**	163.405
Case 4	**PSNR** ↑	13.519	23.665	22.457	25.234	26.217	33.107	32.138	31.893	32.761	24.237	34.497	**34.525**
	SSIM ↑	0.151	0.593	0.597	0.744	0.761	0.876	0.884	0.878	0.904	0.617	**0.907**	0.906
	FSIM ↑	0.548	0.849	0.837	0.920	0.931	0.954	0.947	0.960	0.954	0.871	**0.963**	**0.963**
	ERGAS ↓	955.420	299.723	338.720	252.098	239.614	102.836	111.127	115.648	105.227	291.546	84.347	**84.183**
Case 5	**PSNR** ↑	21.111	30.918	28.135	30.808	32.797	34.061	32.282	32.471	33.159	32.702	**34.646**	34.644
	SSIM ↑	0.374	0.829	0.802	0.857	0.874	0.895	0.893	0.869	0.914	0.870	0.917	**0.919**
	FSIM ↑	0.702	0.941	0.917	0.947	0.955	0.959	0.950	0.957	0.960	0.962	**0.966**	**0.966**
	ERGAS ↓	403.867	130.915	176.698	132.231	117.991	90.769	109.961	112.555	101.451	111.032	83.395	**83.341**
Case 6	**PSNR** ↑	13.362	23.688	22.560	25.023	26.221	32.919	31.544	30.635	32.260	24.275	33.762	**33.784**
	SSIM ↑	0.144	0.585	0.593	0.732	0.753	0.876	0.877	0.862	0.896	0.608	0.907	**0.910**
	FSIM ↑	0.540	0.843	0.833	0.912	0.924	0.951	0.944	0.949	0.951	0.864	**0.961**	**0.961**
	ERGAS ↓	971.929	298.366	334.637	258.858	239.536	106.068	119.506	135.412	112.202	292.045	92.375	**92.058**
Time	**Average**	-	175.6	393.0	642.3	26.0	578.1	64.6	22.1	89.4	**0.6**	111.7	108.1

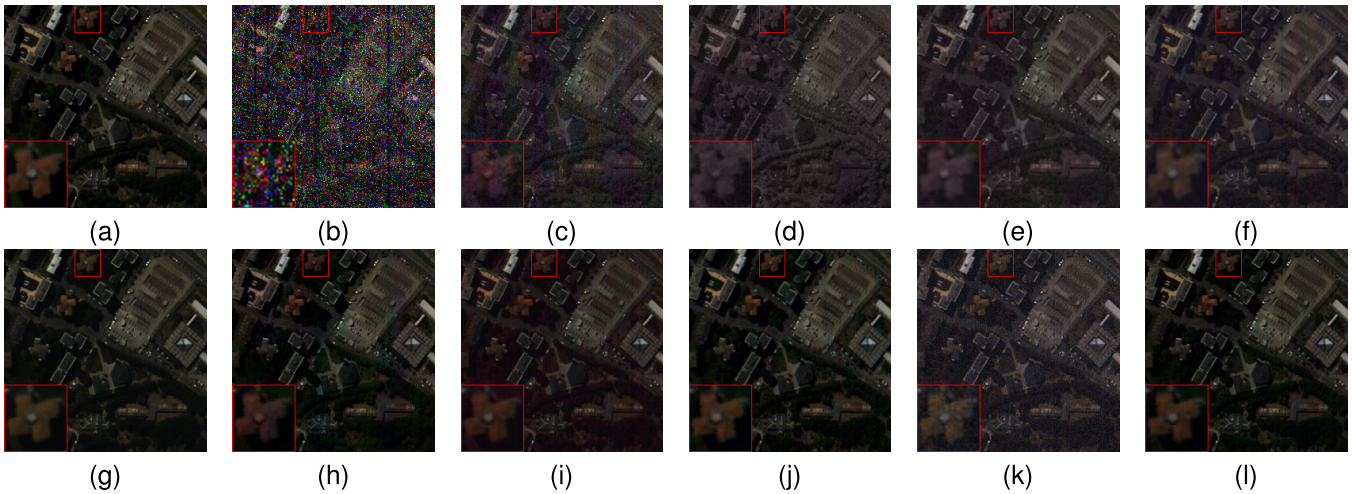


Fig. 12. Denoising results of the Pavia Center. (a) False-color original image with bands (R: 55, G: 35, and B: 15). (b) Noisy image. (c) LRMR. (d) NonLRMA. (e) SMDS. (f) RCILD. (g) WNLRATV. (h) NFF. (i) FastHyMix. (j) 3DCTV. (k) FallHyDe. (l) GNRSR$_{\text{NFF}}$.

220. This dataset is seriously degraded by the mixture of impulse noise, dense water vapor, and atmosphere as shown in Fig. 15(a).

Figs. 16 and 17 respectively show the denoising visual performance of the pseudo-color image of bands 142, 22, and 1 and the gray image for band 1 of the Indian Pines dataset via different methods. For HSI bands severely degraded by Gaussian noise and impulse noise, such as band 1, LRMR cannot effectively remove Gaussian noise and impulse noise. SMDS and NFF excessively remove noise and are unable to restore the original spatial structure. NonLRMA effectively suppresses noise, but some edges are weakened, and the low-rank components cannot be accurately restored. WNLRATV, FastHyMix, and 3DCTV fail to preserve spectral information while eliminating noise, and introduce the color of noise into the denoised image. RCILD, FallHyDe, and the proposed method, whether using NFF or ℓ_p-norm as the specific nonconvex surrogate, are all effective in removing noise and preserving complex image details.

2) Experimental Results on HYDICE Urban Dataset: The HYDICE Urban dataset, acquired by the HYDICE sensor, is a widely used hyperspectral dataset consisting of 307 × 307 pixels and 210 bands. As shown in Fig. 15(b), the dataset is corrupted by atmosphere, impulse noise, deadlines, and stripes.

Figs. 18 and 19 respectively present the denoising results of the pseudo-color image of bands 69, 106, and 151 and the gray image for band 106 of the Urban dataset via different methods. For heavily corrupted HSI bands, such as band 106, LRMR is basically ineffective in removing dense horizontal stripes, while SMDS can only remove part of the dense horizontal stripes. RCILD and FastHyMix successfully remove almost all horizontal stripes, but perform poorly in

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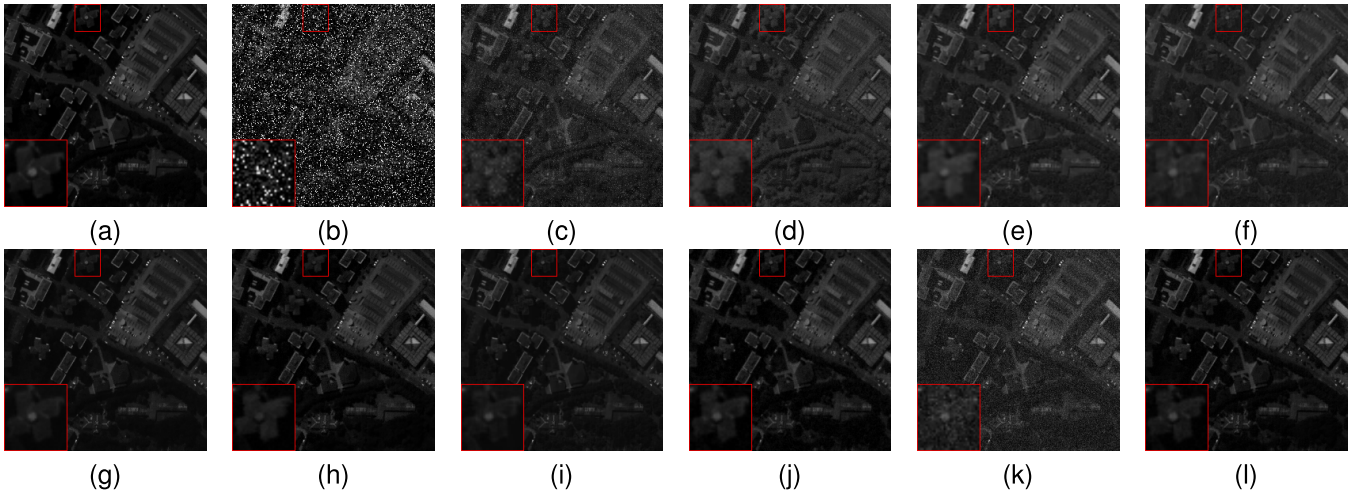


Fig. 13. Denoising results of the Pavia Center. (a) Original image with band 15. (b) Noisy image. (c) LRMR. (d) NonLRMA. (e) SMDS. (f) RCILD. (g) WNLRATV. (h) NFF. (i) FastHyMix. (j) 3DCTV. (k) FallHyDe. (l) GNRSR$_{\text{NFF}}$.

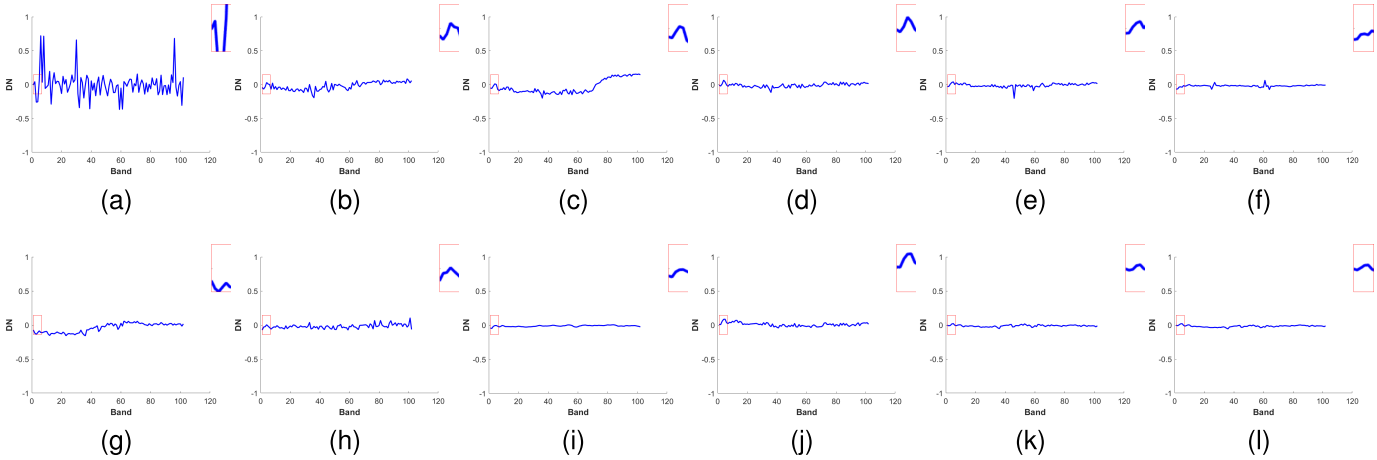


Fig. 14. Difference between the spectrum of the noise-free pixel (17, 103) and the corresponding restored pixels by different methods in the Pavia Center image in Case 6. (a) Noisy. (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

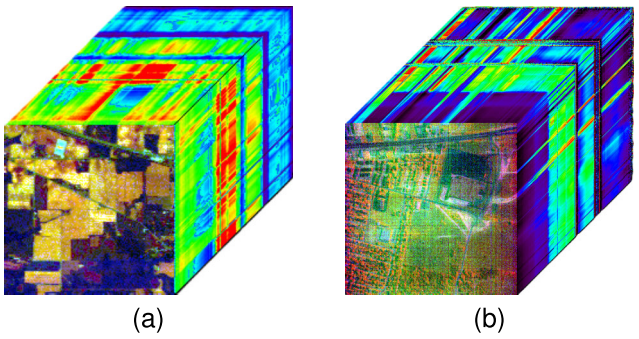


Fig. 15. (a) Degraded AVIRIS Indian Pines dataset. (b) Degraded HYDICE Urban dataset.

removing vertical stripes. WNLRATV significantly eliminates most of the mixed noise, but, unfortunately, oversmooths some image details and spectral signatures are destroyed. FallHyDe effectively removes all noise and preserves image details well, but it results in severe spectral distortion. NonLRMA, NFF, GNRSR$_{\text{NFF}}$, and GNRSR$_{\ell_p}$ all exhibit superior denoising performance and achieve the best visual outcomes.

In summary, through quantitative and visual assessments, the proposed GNRSR method surpasses all competitors in terms of mixed noise removal, texture detail restoration, and spectral preservation.

V. Discussion

In this section, we briefly discuss the impact of several important parameters in the model on the denoising performance and the choice of the forward transform operator.

A. Parameter Analysis

In our proposed methods GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$, the denoising performance is affected by the regularization parameters λ and γ, the adjustable parameters α and p, and the noise-bound parameter δ. We select the Pavia Center dataset under Case 6 to perform a sensitivity analysis on GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$.

1) Parameters λ and γ: The regularization parameters λ and γ balance the weight among the generalized nonconvex function $\Phi(X)$, the sparse noise term $\|S\|_1$, and the low-rank component sparse term $\|\mathcal{G}(X)\|_1$. By fixing the value of α, we

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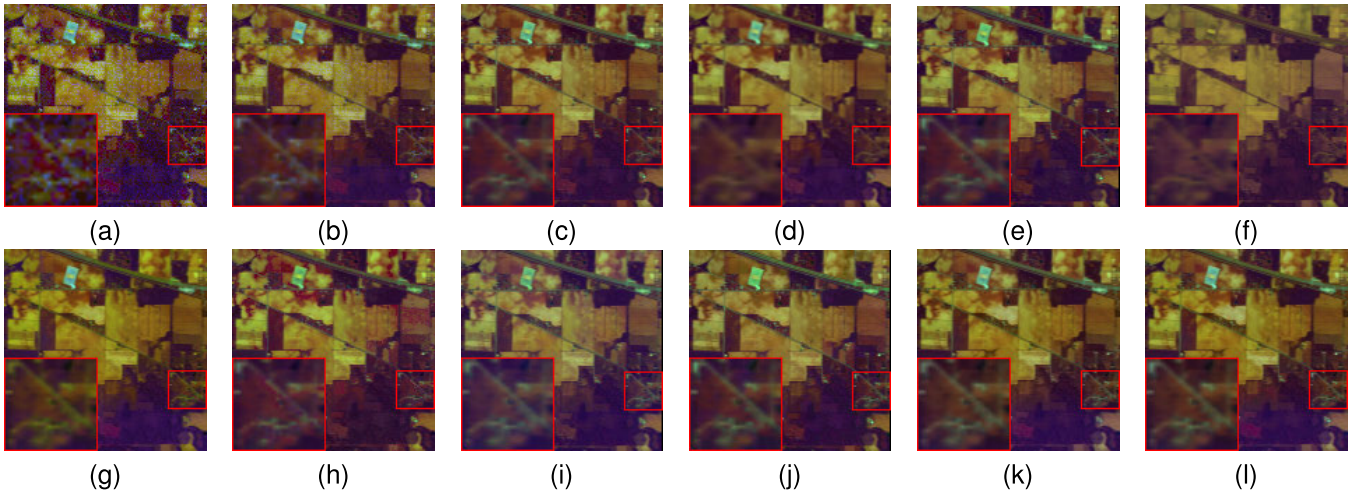


Fig. 16. Denoising results of the AVIRIS Indian Pines dataset. (a) False-color original image with bands (R: 142, G: 22, and B: 1). (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

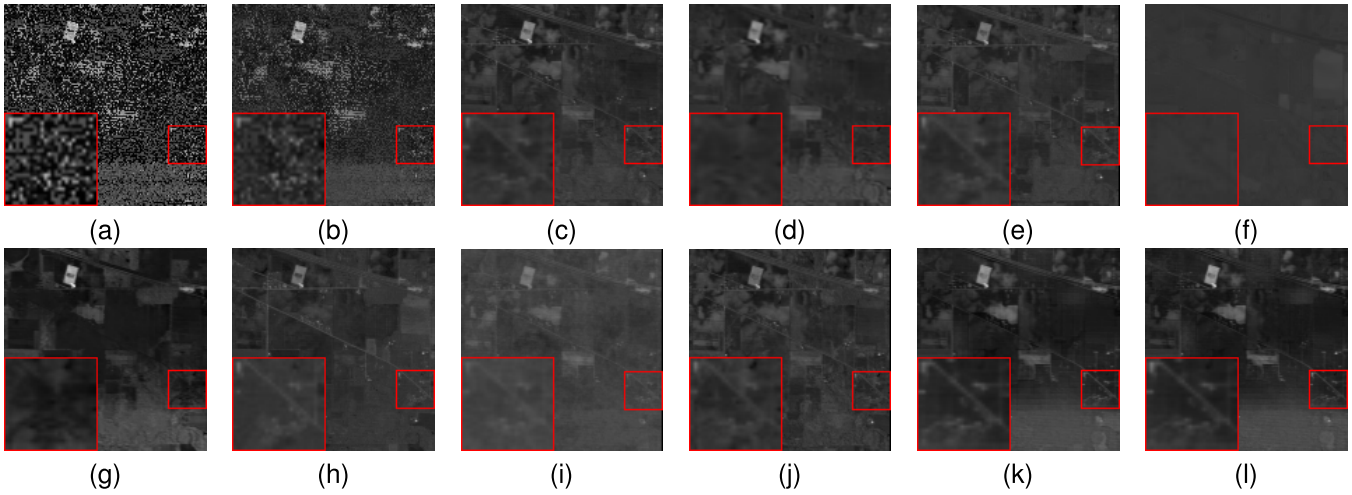


Fig. 17. Denoising results of the AVIRIS Indian Pines dataset. (a) Original image with band 1. (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

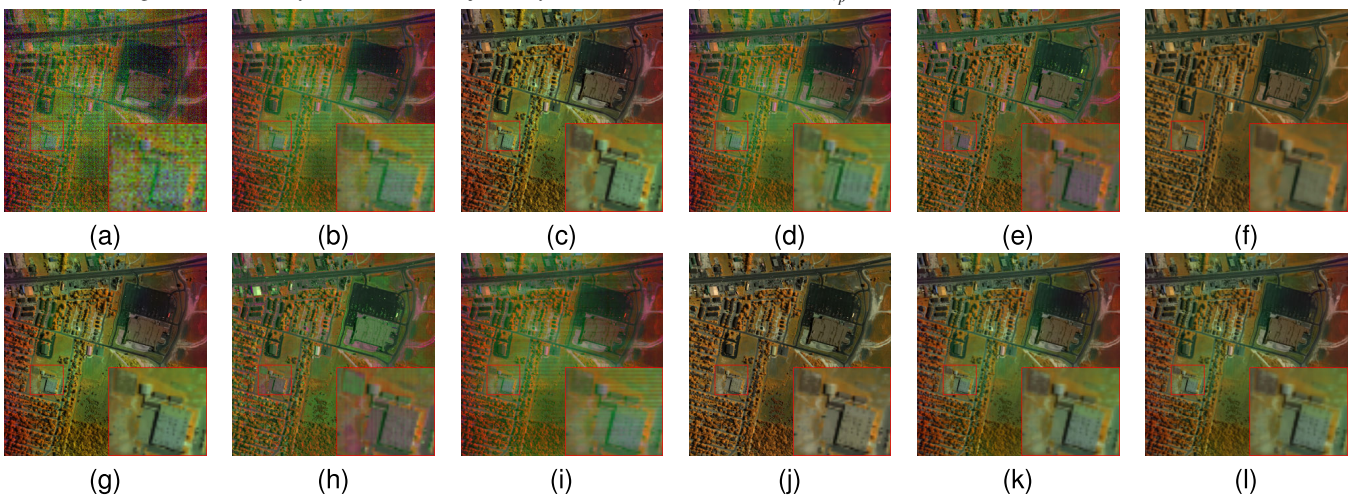


Fig. 18. Denoising results of the HYDICE Urban dataset. (a) False-color original image with bands (R: 71, G: 107, and B: 207). (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

vary λ within the range of [0.003, 0.019] and γ within the range of [0.025, 0.055] for GNRSR$_{\text{NFF}}$. For GNRSR$_{\ell_p}$, the value of p is fixed, and λ is kept within the same range, but γ is adjusted to fall between [0.04, 0.10]. Fig. 20 demonstrates the PSNR and SSIM performance of both GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$ as λ and γ vary. It can be observed that the PSNR

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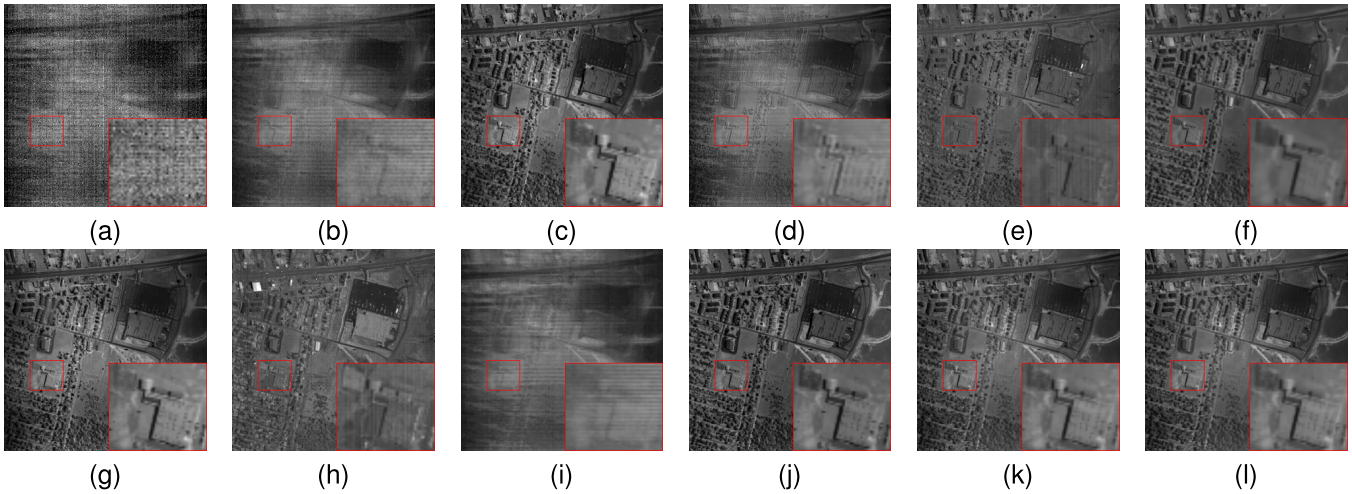


Fig. 19. Denoising results of the HYDICE Urban dataset. (a) Original image with band 107. (b) LRMR. (c) NonLRMA. (d) SMDS. (e) RCILD. (f) WNLRATV. (g) NFF. (h) FastHyMix. (i) 3DCTV. (j) FallHyDe. (k) GNRSR$_{\text{NFF}}$. (l) GNRSR$_{\ell_p}$.

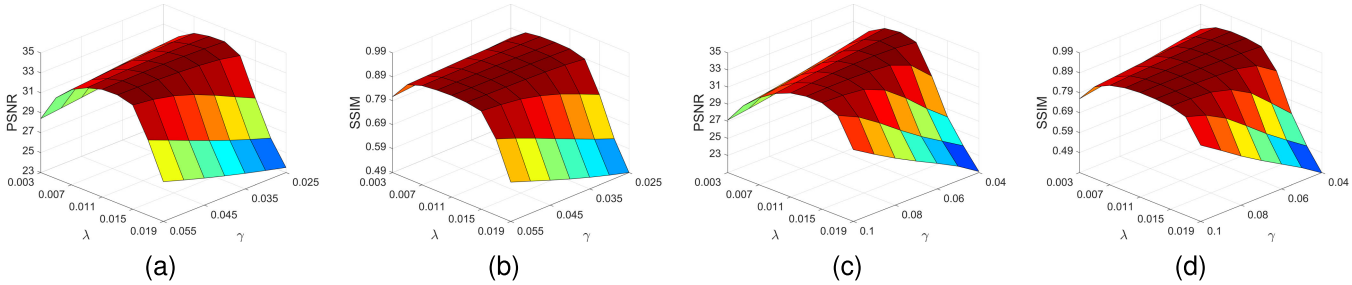


Fig. 20. PSNR and SSIM values change of GNRSR$_{\text{NFF}}$ and GNRSR$_{\ell_p}$ with respect to parameters λ and γ. (a) PSNR value changes of GNRSR$_{\text{NFF}}$. (b) SSIM changes of GNRSR$_{\text{NFF}}$. (c) PSNR value changes of GNRSR$_{\ell_p}$. (d) SSIM value changes of GNRSR$_{\ell_p}$.

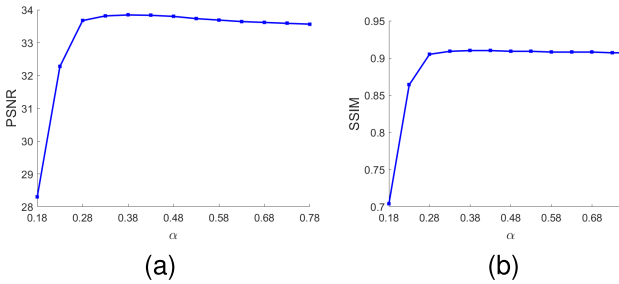


Fig. 21. PSNR and SSIM values change of GNRSR$_{\text{NFF}}$ with parameter α. (a) PSNR value changes of GNRSR$_{\text{NFF}}$. (b) SSIM value changes of GNRSR$_{\text{NFF}}$.

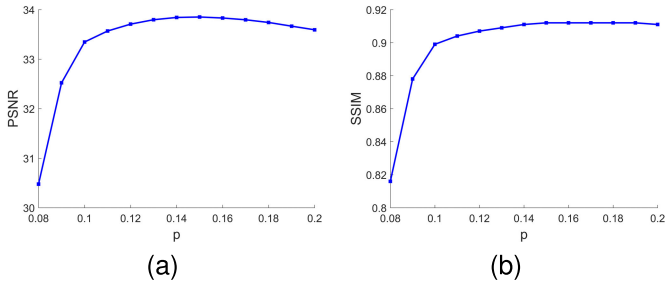


Fig. 22. PSNR and SSIM values change of GNRSR$_{\ell_p}$ with parameter p. (a) PSNR value changes of GNRSR$_{\ell_p}$. (b) SSIM value changes of GNRSR$_{\ell_p}$.

values of the proposed GNRSR$_{\text{NFF}}$ remain stable when λ is in the range [0.009, 0.013] and γ in the range [0.025, 0.045], while the SSIM values are stable when λ is in the range [0.007, 0.013] and γ in the range [0.025, 0.055]. For the GNRSR$_{\ell_p}$ method, the PSNR values exhibit stability when λ is in the range [0.007, 0.011] and γ in the range [0.04, 0.06], while the SSIM values remain stable when λ is in the range [0.007, 0.011] and γ in the range [0.04, 0.1]. Therefore, in our experiments, we set $\lambda = 0.011$ and $\gamma = 0.035$ for GNRSR$_{\text{NFF}}$, and $\lambda = 0.009$ and $\gamma = 0.04$ for GNRSR$_{\ell_p}$.

2) Parameters α and p: The parameter α in GNRSR$_{\text{NFF}}$ controls the degree of contraction of the singular values, while the parameter p in GNRSR$_{\ell_p}$ determines the shrinkage level of singular values. With λ and γ set to their optimal values, α is adjusted within the range [0.18, 0.78] for GNRSR$_{\text{NFF}}$, and p is varied within the range [0.08, 0.20] for GNRSR$_{\ell_p}$. Fig. 21 illustrates the relationship between PSNR and SSIM performance and the parameter α of the GNRSR$_{\text{NFF}}$ method. As can be seen, the PSNR and SSIM values gradually increase as α rises within the range [0.18, 0.38], then slowly decline as α continues to increase. To achieve optimal performance, we set $\alpha = 0.38$ for GNRSR$_{\text{NFF}}$ in our experiments. Similarly, Fig. 22 depicts the variation of PSNR and SSIM with p for the GNRSR$_{\ell_p}$ method. The PSNR and SSIM values increase gradually as p grows within the range [0.08, 0.15], before slowly decreasing as p increases further. Thus, we set $p = 0.15$ for GNRSR$_{\ell_p}$ in our experiments to achieve the best performance.

3) Noise-Bound Parameter δ: The parameter δ controls the allowable residual energy in the constraint $\|Y-X-S\|_F \leq \delta$. By fixing other parameters, δ is varied over a wide range of values, and the corresponding PSNR and SSIM results are

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TABLE IV
PSNR AND SSIM VALUES CHANGE OF $\text{GNRSR}_{\text{NFF}}$ WITH PARAMETER δ

Dataset	Indexes	$\delta = 10^{-1}$	$\delta = 10^{-2}$	$\delta = 10^{-3}$	$\delta = 10^{-4}$	$\delta = 10^{-5}$	$\delta = 10^{-6}$	$\delta = 10^{-7}$
WDC (Case 6)	**PSNR** ↑	34.553	34.917	34.940	34.945	**34.946**	**34.946**	**34.946**
	SSIM ↑	0.953	**0.955**	**0.955**	**0.955**	**0.955**	**0.955**	**0.955**
	Time ↓	**123.3**	172.0	215.3	264.1	305.8	352.9	398.9
Pavia (Case 6)	**PSNR** ↑	33.434	34.006	34.041	34.046	**34.048**	**34.048**	**34.048**
	SSIM ↑	0.903	0.916	**0.917**	**0.917**	**0.917**	**0.917**	**0.917**
	Time ↓	**53.9**	72.8	92.5	111.7	132.4	148.8	167.4

TABLE V
DENOISING RESULTS OF CASE 6 ON THE PAVIA CENTER DATASET USING DIFFERENT FORWARD TRANSFORM OPERATORS

Indexes	Noised	DFT	HT	DCT
PSNR ↑	13.362	32.886	32.903	**33.762**
SSIM ↑	0.144	0.893	0.896	**0.907**
FSIM ↑	0.540	0.955	0.956	**0.961**
ERGAS ↓	971.929	102.513	102.440	**92.375**

TABLE VI
ABLATION STUDY OF THE GNRSR MODEL

Dataset	Indexes	GNRSR-S	GNRSR-G	GNRSR
WDC (Case 6)	**PSNR** ↑	33.461	30.748	**34.945**
	SSIM ↑	0.937	0.895	**0.955**
	FSIM ↑	0.963	0.942	**0.974**
	ERGAS ↓	83.003	115.596	**71.639**
Pavia (Case 6)	**PSNR** ↑	31.969	30.415	**34.046**
	SSIM ↑	0.886	0.865	**0.917**
	FSIM ↑	0.947	0.931	**0.964**
	ERGAS ↓	116.498	134.193	**91.019**

reported in Table IV. It can be observed that the performance of $\text{GNRSR}_{\text{NFF}}$ remains stable as δ changes, indicating that the proposed GNRSR model is insensitive to this parameter. Therefore, in all our experiments, we set $\delta = 1 \times 10^{-4}$ for both $\text{GNRSR}_{\text{NFF}}$ and GNRSR_{ℓ_p}.

B. Forward Transform Operator Analysis

Since the matrices obtained through different forward transform operators are different, the amount of information that can be retained about the sparse information of the matrices is also different, and therefore the corresponding denoising performance is also different. Table V shows the denoising results of the $\text{GNRSR}_{\text{NFF}}$ method on the Pavia Center dataset under Case 6, in which three different forward transform operators are used, namely DFT operator, HT operator, and DCT operator. It is evident that the $\text{GNRSR}_{\text{NFF}}$ method achieves comparable performance when using the DFT and HT operators as the forward transform operators. However, the $\text{GNRSR}_{\text{NFF}}$ method achieves the best denoising performance when using the DCT operator. Accordingly, we recommend selecting DCT operator as the forward transform operator.

C. Ablation Study

In the proposed GNRSR model (12), the clean HSI X is simultaneously characterized by two regularization terms, i.e., a generalized nonconvex surrogate and a sparse regularizer. To investigate their individual contributions, ablation experiments are carried out. For the baseline GNRSR, the following two models can be generated by it:

$$\min_{X,S} \ \Phi(X) + \lambda\|S\|_1 \quad \text{s.t.} \quad \|Y-X-S\|_F \le \delta \tag{32}$$

$$\min_{X,S} \ \lambda\|S\|_1 + \gamma\|\mathcal{G}(X)\|_1 \quad \text{s.t.} \quad \|Y-X-S\|_F \le \delta. \tag{33}$$

The models in (32) and (33) are respectively denoted as "GNRSR-S" and "GNRSR-G." In particular, in both the GNRSR and GNRSR-S models, we adopt the NFF penalty as a specific instantiation of the generalized nonconvex low-rank regularizer. In the GNRSR and GNRSR-G models, we employ the DCT operator as the forward transform operator. All models are implemented with their own optimal parameter settings, while all the other settings, including initialization, are kept the same for fairness.

The most challenging cases on the two datasets, i.e., WDC (Case 6) and Pavia Center (Case 6), are tested. The quantitative results of the ablation experiments are reported in Table VI. We have the following observations.

1) Table VI shows that, compared with GNRSR-S, GNRSR achieves consistent denoising performance on both datasets. This demonstrates that introducing the sparse regularizer ℓ_1 for X helps GNRSR better preserve fine details.
2) Table VI also shows that, compared with GNRSR-G, GNRSR significantly improves denoising performance on both datasets. This indicates that incorporating the generalized nonconvex rank prior enables GNRSR to better discriminate the clean HSI from noise, while still maintaining the advantages of the transform-domain sparsity.

Hence, the generalized nonconvex surrogate and the sparse regularizer provide complementary contributions: the former mainly improves spectral coherence, while the latter mainly enhances spatial structures.

VI. CONCLUSION

In this article, the GNRSR model was proposed for HSI denoising. The GNRSR model had two distinct advantages.

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First, it could effectively capture complex details such as textures and edges in clean HSI. Second, it could flexibly adopt various nonconvex low-rank regularization terms for specific problem to enhance its denoising ability. To effectively solve the proposed model, we presented an accurate algorithm based on inexact ALM framework with theoretical guarantees. The effectiveness of GNRSR method in dealing with Gaussian noise and mixed noise was proved by simulation data and real data experiments, and the experimental results showed that its denoising performance was superior to many state-of-the-art HSI denoising methods.

Despite the effectiveness of the proposed method, there is still room for further improvement. Recently, HSI has been increasingly modeled as a third-order tensor [56], [57], [58], and tensor-based methods offer notable theoretical advantages in terms of energy conservation and structural integrity during decomposition. Building on this insight, our future work will focus on extending the study to low-rank tensor approximation, thereby enabling better preservation and exploitation of the intrinsic structural information and prior knowledge of HSIs.

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Yusen Tan received the B.S. degree in computer science from Southwest University, Chongqing, China, in 2025. He is currently pursuing the M.S. degree in data science and analytics with The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China.

His research focuses on machine learning.



Dong Hu received the B.S. and M.S. degrees in computer science from Southwest University, Chongqing, China, in 2013 and 2018, respectively, where she is currently pursuing the Ph.D. degree in computer science with the College of Computer and Information Science.

Her research interests include machine learning and convex/nonconvex optimization.



Yao Fu is currently pursuing the M.S. degree in computer science with the College of Computer and Information Science, Southwest University, Chongqing, China.

Her research interests include machine learning and data mining.



Chunming Yang received the B.S. degree in computer science from Southwest University, Chongqing, China, in 2001, and the M.S. degree in computer application technology from Southwest University of Science and Technology, Mianyang, China, in 2011.

He is currently a Professor with the School of Computer Science and Technology, Southwest University of Science and Technology. His research interests include machine learning, data mining, and natural language processing.



Zhi Wang (Member, IEEE) received the B.S. and M.S. degrees in computer science, and the Ph.D. degree in statistics from the Southwest University, Chongqing, China, in 2005, 2009, and 2020, respectively.

He is currently an Associate Professor with the College of Computer and Information Science, Southwest University. He has published more than 30 academic articles in reputable journals and conferences, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON SIGNAL AND INFORMATION PROCESSING OVER NETWORKS, and IJCAI. His research interests include machine learning, data mining, and convex/nonconvex optimization.

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